

DSA02 COMPUTER VISION WITH OPENCV L T P C 3 0 2 4 Prerequisite: CSA08

Python Programming

Course Objectives

The course on Computer Vision with OpenCV aims to provide students with the following:

1. To implement fundamental image processing techniques required for computer vision and Understand Image formation process
2. To perform shape analysis and extract features from Images and do analysis of Images
3. Generate 3D model from images
4. To develop applications using computer vision techniques
5. Understand video processing, motion computation and 3D vision and geometry

Course Outcomes

On successful completion of the course, the student will be able to:

1. Implement fundamental image processing techniques required for computer vision and understand the image formation process.
2. Perform shape analysis - extract features from images - analyze images.
3. Generate 3D models from images.
4. Develop applications using computer vision techniques.
5. Understand video processing - motion computation - 3D vision - geometry.

Topics

Introduction to Computer Vision and Image Formation

Image Processing, Computer Vision, and Computer Graphics - Introduction to Computer Vision - Low-level, Mid-level, High-level Vision - Overview of Diverse Computer Vision Applications: Document Image Analysis, Biometrics, Object Recognition, Tracking, Medical Image Analysis, Content-Based Image Retrieval, Video Data Processing, Multimedia, Virtual Reality, and Augmented Reality - Camera Models and Image Formation - Camera Model and Camera Calibration - Binocular Imaging Systems - Multiple View Geometry - Structure Determination - Shape from Shading - Photometric Stereo - Depth from Defocus - Construction of 3D Model from Images

Image Processing and Feature Extraction

Image Preprocessing - Image Representations (Continuous and Discrete) - Edge Detection - Feature Extraction Techniques - Motion Estimation: Regularization Theory, Optical Flow Computation, Stereo Vision, Motion Estimation, Structure from Motion

Shape Representation

Shape and Object Representation - Contour-Based Representation - Region-Based Representation - Deformable Curves and Surfaces - Snakes and Active Contours -

Level Set Representations - Fourier and Wavelet Descriptors - Medial Representations - Multi-Resolution Analysis

Recognition and Learning in Vision

Hough Transforms and Other Object Recognition Methods - Shape Correspondence and Shape Matching - Principal Component Analysis - Shape Priors for Recognition - Deep Learning for Image Processing - Convolutional Neural Networks (CNNs) - Transformers for Vision - Generative Adversarial Networks (GANs) - Pattern Recognition and Statistical Methods - Pattern Recognition Methods - Hidden Markov Models (HMM) - Gaussian Mixture Models (GMM) - Expectation-Maximization (EM) - Support Vector Machines (SVM) - Graph-Based Learning in Vision

Applications

Photo Album Organization - Face Detection - Face Recognition - Eigenfaces - Active Appearance and 3D Shape Models of Faces - Surveillance: Foreground-Background Separation - Particle Filters - Chamfer Matching - Tracking and Occlusion Handling - Combining Views from Multiple Cameras - Human Gait Analysis - Advanced Applications - In-Vehicle Vision System: Locating Roadways - Road Markings - Identifying Road Signs - Locating Pedestrians - Autonomous Vehicle Perception - Medical Image Segmentation - Augmented Reality and Mixed Reality.

Text Books

1. Computer Vision - A modern approach, by D. Forsyth and J. Ponce, Prentice Hall
Robot Vision, by B. K. P. Horn, McGraw-Hill.
2. Introductory Techniques for 3D Computer Vision, by E. Trucco and A. Verri,
Publisher: Prentice Hall.

References

1. R. C. Gonzalez, R. E. Woods. Digital Image Processing. Addison Wesley Longman, Inc., 1992.
2. D. H. Ballard, C. M. Brown. Computer Vision. Prentice-Hall, Englewood Cliffs, 1982.
3. Richard Szeliski, Computer Vision: Algorithms and Applications (CVAA). Springer, 2010 Image Processing, Analysis, and Machine Vision. Sonka, Hlavac, and Boyle. Thomson.
4. E. R. Davies, Computer & Machine Vision, Fourth Edition, Academic Press, 2012
5. Simon J. D. Prince, Computer Vision: Models, Learning, and Inference, Cambridge University Press, 2012
6. Mark Nixon and Alberto S. Aquado, Feature Extraction & Image Processing for Computer Vision, Third Edition, Academic Press, 2012.